

Exam 2026: Household behavior over the lifecycle

KU-identification: lhr250

Q1

We have two state variables in the model, b , which is the birth cohort and k_t , which is the level of human capital. A variable only needs to be a separate state if it carries information that is not already contained in the other states. The year y_t is given as $y_t = t + b$, and is simply the age plus birth cohort. Both of these are already in the model as we can find the age t as it is the time dimension we solve over by backward induction, whereas the birth cohort b is a state, because it is fixed for each individual through their entire life. So once we know the person's age and their cohort, we can calculate the year.

We solve the model age by age, starting from the end of life. In the final period there is no future left, so the optimal choice is easy to find. Knowing the value of reaching that last period, we step back one age and solve the period before it, using the answer we just found. We repeat this all the way back to age $t = 0$. We go backwards because each period's best choice depends on what comes after it, and by starting at the end we always already know the future when we solve the present.

The year matters only because it scales wages through the aggregate process $\lambda(y_t) = 1 + \lambda_0 y_t$. But since y_t is fully pinned down by (t, b) , we can compute the wage directly from the two states we already track. Adding y_t as its own state would just duplicate information we already have. That is why we do not keep the year as a separate state.

Q2

2.a)

Figure 2a illustrates the labor supply for the two models. There are two counterintuitive effects in this figure. First, the labor supply declines with age in both models. Second, the figure shows how workers supply more labor under the high tax ($\tau = 0.3$) than under no tax ($\tau = 0$) at every age. Both of these characteristics are a result of the fact that the CRRA coefficient: $\rho = 1.5 > 1$.

To break it down, we start by looking at the consumption function, ignoring the continuation value for a moment: $c_t = (1 - \tau)w_t l_t$. The first-order condition gives:

$$l_t^* = \left[\frac{[(1 - \tau)w_t]^{1-\rho}}{\phi} \right]^{\frac{1}{\rho+\eta}}$$

With $\rho = 1.5$ and $\eta = 2.0$ the exponent is: $\frac{1-\rho}{\rho+\eta} = \frac{-0.5}{3.5} < 0$. Since the exponent is negative, working hours move opposite to the net wage. So when the tax lowers the net wage, working hours rise to maintain consumption.

When the net wage changes, two forces pull working hours in opposite directions. The substitution effect states that a lower net wage makes leisure cheaper, why you should work less. There is also an income effect, which states that a lower net wage makes you poorer, why you should

work more to maintain consumption. The sign of the exponent tells us which force wins. Since the exponent is negative the income effect dominates. As $\rho > 1$ the desire to keep consumption smooth is strong, so the income effect outweighs the substitution effect. This has two consequences. The tax lowers the net wage, so workers work more to maintain consumption, which is why the model including the tax is above the one with no tax. Human capital accumulation raises the wage over the life cycle, so workers grow richer and work less, which explain the downward-sloping graph.

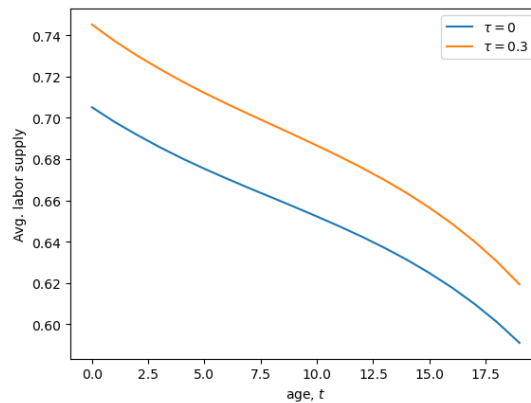


Figure 2a: Age profiles of avg. labor supply for $\tau = 0$ and $\tau = 0.3$

2.b)

Figure 2b shows the employment rate for the two models. The two graphs lie exactly on top of each other and is equal to one, indicating that everyone chooses to work in both models. The employment rate stays the same across all age profiles. This illustrates how no one chooses not to work, despite of the tax being introduced. The reason is that the only alternative to working is receiving unemployment benefits of $\pi = 0.5$, which is small compared to what people can earn. Even under the high tax and at the lowest levels of human capital, working the optimal number of hours gives higher total utility than staying at home receiving the benefit of $\pi = 0.5$. So not working $l_t = 0$ is never optimal. The tax lowers the reward to working but leaves the unemployment benefit unchanged. But the wage is so much larger than π_t , that no one chooses to decide not to work. Figure 2a illustrates how the income effect pushes working hours up. This tells us that in this model the tax works entirely through how much people work and not whether or not they work.

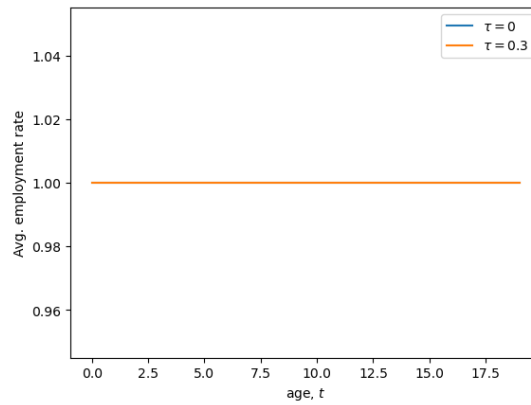


Figure 2b: Age profiles of avg. employment rate for $\tau = 0$ and $\tau = 0.3$

2.c)

Figure 2c illustrates the average consumption over all age profiles. We remember that there are no savings in this model, so people cannot move money between periods. So, what you earn in a given period is what you consume. The model including the tax sits below the model with no tax at every age, showing that the consumption is lower under the high tax, $\tau = 0.3$ than under no tax. This is because the tax takes a share of the earnings, so people have less money to spend. As figure 2a illustrated people work more when taxed, but working more cannot make up for giving up part of your income to the government, so consumption still ends up lower.

Both graphs rise when young and fall when older. When you are young, human capital builds up fast so wages increase quickly, and even though working hours decline with age, the higher wage result in higher income, which is reflected in higher consumption. Later in life human capital does not grow as much anymore, why wages flatten and working hours decline, resulting in the consumption falling.

The hump shape reflects the missing savings. If people were able to save, they would borrow when young and set money aside in mid-life to keep consumption smooth through the entire life cycle. Without savings, consumption has no choice but to follow the earnings.

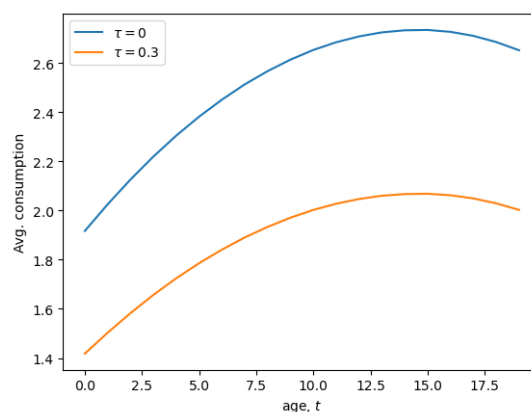


Figure 2c: Age profiles of avg. consumption for $\tau = 0$ and $\tau = 0.3$

Q3

3.a)

At the baseline policy $\theta = (\pi_0 = 0.5, \tau = 0.3)$ the government budget is

$$G_3 = \sum_{i=1}^N \sum_{t=1}^T G_{i,t} \approx 808,285.$$

G_3 is the sum over all $N = 50,000$ individuals and all $T = 20$ periods. The positive and large G_3 shows that the government runs a large surplus. That means that the labor-income tax it collects from workers are larger than the unemployment benefits it pays out to people who do not work (but as no one is working there is nothing subtracted).

Note. The exam lists the cohort wage constants as $(0.7, 0.8, 0.9, 1.1, 1.2)$, but the baseline code (.py file) sets them using `np.linspace(0.7, 1.2, Nb) = (0.7, 0.825, 0.95, 1.075, 1.2)`. I keep the baseline code as is and use them throughout, so G_3 and all later comparisons stay consistent. Throughout the rest of the exam answers, year and age indices start at 0, so a year labelled y in the figures corresponds to the exam's year $y + 1$ (and likewise for age).

3.b+c)

Figure 3b illustrates a Laffer curve showing how the government budget develops as the tax rate increases. The plot is hump-shaped and peaks around a tax-rate of $\tau \approx 0.7$ and then declines sharply. It illustrates how the labor supply reacts to changes in the tax-rate. The higher the tax rate, the smaller the share a worker gets to keep of his earned wage. Raising the tax-rate, τ pulls government revenue, $\tau w_t l_t$ two ways at once. Mechanically, the government takes a larger share of each unit earned, which increases revenue. But behaviorally, a lower net wage makes work less rewarding, so people choose to work less, which result in the tax base, $w_t l_t$, falling, which leads to lower revenue. At lower levels of τ the mechanical force wins and the budget increases. At high levels of τ , the behavioral force dominates and the budget falls drastically.

The fall is sharp, as work becomes barely worth it a very high tax-rates. Some people respond by cutting hours, while others stop working entirely. Each person who quits working costs the government twice, as there are both lost taxes and they also have to pay unemployment benefits. This is also why the revenue can end up negative at the highest rates, reflecting a point where no one is working and everyone receives unemployment benefits. In addition, wages are built through by accumulating human capital, why cutting hours today lowers future wages and therefore also the future tax base as well. So the government revenue loss compounds over the life cycle.

The baseline $\tau = 0.3$ is on the side of the Laffer curve where higher taxes still raise government revenue. However maximizing revenue with a tax-rate of $\tau = 0.7$ is not a desirable policy. It depends largely on how much people dislike work, and a rate that high would end up crushing both work and welfare.

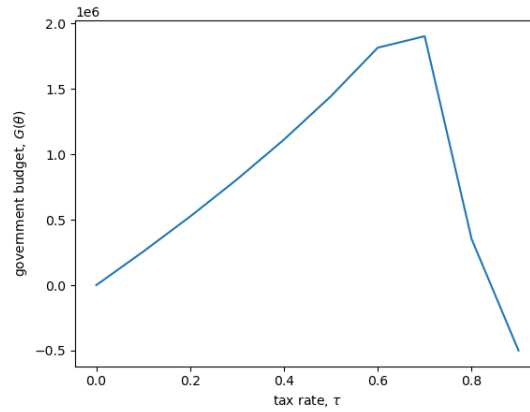


Figure 3b: Laffer curve, $G(\theta)$

Q4

4.a)

The reform is an extra unemployment benefit restricted on age as well as year. It affects the unemployment benefit $\pi_t = \pi_0$, which now needs to include the extra benefit, π_1 . As it is only given to individuals who are 10 years or older, we set $\mathbf{1}\{t \geq 10\}$, and since it is first starting from year 8 and onward, we set $\mathbf{1}\{y_t \geq 8\}$:

$$\pi_t = \pi_0 + \pi_1 \mathbf{1}\{t \geq 10\} \mathbf{1}\{y_t \geq 8\}, \quad y_t = t + b,$$

with $\pi_1 = 0.6$. The budget constraint looks the same, but is affected by the changes in π_t . $c_t = (1 - \tau)w_t l_t + \pi_t \mathbf{1}\{l_t = 0\}$. A person not working now ends up receiving $\pi_0 + \pi_1 = 1.1$, if he fulfills the criteria. The Bellman equation, the human-capital equation, the wage equation and the terminal period are unchanged.

The reform is anticipated automatically by individuals in the model because we use backward induction to solve the model. Using backward induction we solve from $t = T$ and downward, with the added unemployment benefit already in place at every period. So when we compute the value at some early period (before the worker is eligible, or before year 8), the continuation value $\mathbb{E}_t[V_{t+1}]$ already contains the future, larger benefits as the later periods were solved first. The worker's choice today therefore reflects the knowledge that the subsidy is coming. This is an example of perfect foresight in a backward-induction solve and requires no special implementation.

4.b)

Figure 4b shows average labor supply and the share working by age, for the baseline and the reform. In Python we simply added the old-age subsidy to the parameters, adjusted some of the functions and then solved and simulated both models and plotted the two outcomes side by side.

The exam counts age from 1: $t \in \{1, 2, \dots, T\}$, so a worker's first period of life is age 1. The code counts from 0, as Python's `range(par.T)` starts at 0, so the same first period of life is code $t = 0$. The exam's "aged 10 or older" and "from year 8" therefore correspond to the thresholds $t \geq 9$ and $y_t \geq 7$ in the code. So for the rest of the exam set, what the figures label year y corresponds to the exam's year $y + 1$ (and likewise for age).

For the first part of life the two models are almost identical. The subsidy is not yet available, so there is nothing to react to and behavior is unchanged. The share working stays at exactly one through age 8 and only falls from age 9, the first eligible age, when the first workers retire. Labor supply moves a bit earlier, as it already is a bit under the baseline at age 8, one period before anyone can retire. This earlier dip is the anticipation effect, where workers cut their hours slightly in advance, as the human capital they would build will no longer be used.

In the baseline everyone works at every age, why the share working stays flat at one, just as in Figure 2b. The reform changes this pattern. As an older unemployed receives $\pi_0 + \pi_1 = 1.1$, which is now more attractive than what they would earn after tax, they stop working altogether. The reform therefore switches on the extensive margin that was completely inactive before, why the question now is no longer only how much people work, but whether they work at all.

The gap keeps widening with age. The first to leave the job market are the low-wage workers, for whom the unemployment benefit most easily beats their pay. As people get older more people leave the market, as there are fewer years left in which working still pays off. The fall in average labor supply comes almost entirely from people leaving work altogether rather than from those still employed working fewer hours. The average share working drops as the pool now contains retirees sitting at zero, which drags the average down.

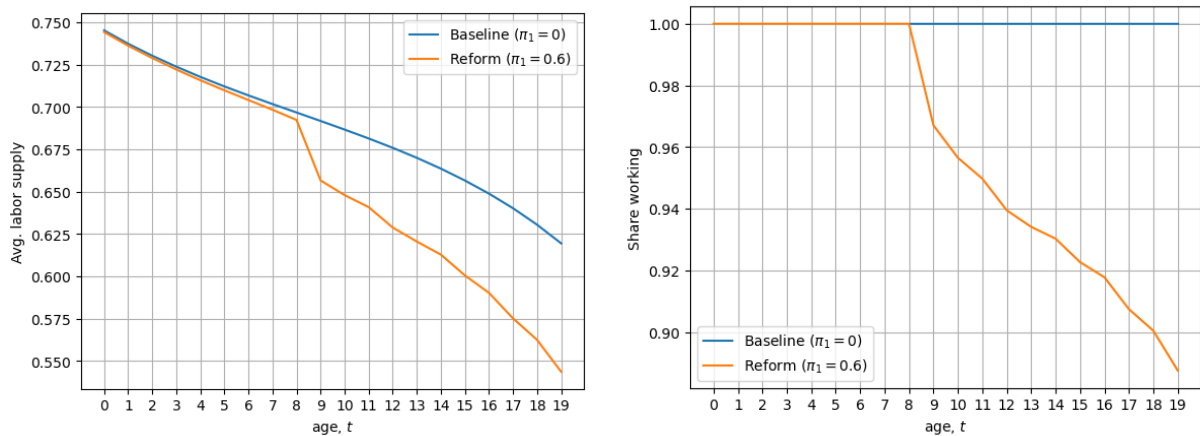


Figure 4b: Age profiles of avg. labor supply and share working

4.c)

We average the stored $G_{i,t} = \tau w_{i,t} l_{i,t} - \pi_t \mathbf{1}\{l_{i,t} = 0\}$ within each calendar year $y = t + b$ for both models. Since the subsidy term includes π_1 , the reform's budget already reflects the extra benefits paid.

The two models coincide until the first workers become eligible to receive the benefit, after which the reform's budget falls below the baseline. At this point (year 8) each retiree stops paying labor tax and starts drawing the benefit, so revenue falls and spending rises for the government at once. This gap is driven by the low-wage cohorts, who are the ones that benefit from accepting the subsidy. The gap then closes again at the very end. This is a composition effect: only old workers from the high cohorts reach the late years (since $y = t + b$), and these are the high earners who keep working even with the subsidy, so almost no retirees are left to draw down the budget at this point.

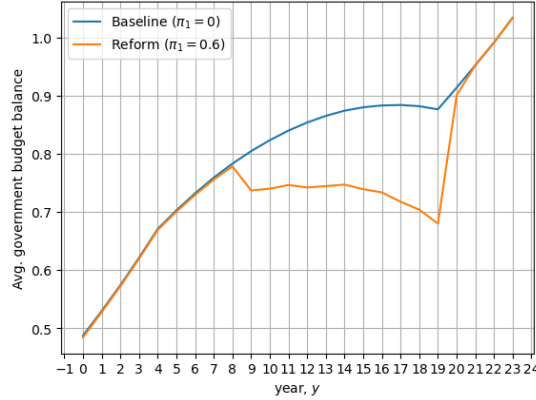


Figure 4c: Average government budget balance across years

Q5

5.a)

As the reform from question 4 drained the government budget, the government decides to add an extra tax from year 8 and onward to finance the old-age retirement reform. The full policy vector is now $\theta = (\pi_0, \pi_1, \tau, \tau_1)$.

The tax is year-targeted, which means that there is no age condition, unlike the subsidy. Everyone working in year 8 or later therefore needs to pay. So the effective tax rate becomes year-dependent:

$$\tau_t = \tau + \tau_1 \mathbf{1}\{y_t \geq 8\}, \quad y_t = t + b,$$

Which changes the budget constraint, which becomes:

$$c_t = (1 - \tau - \tau_1 \mathbf{1}\{y_t \geq 8\}) w_t l_t + \pi_t \mathbf{1}\{l_t = 0\}.$$

The government budget per person is also affected by the new tax:

$$G_{i,t} = (\tau + \tau_1 \mathbf{1}\{y_t \geq 8\}) w_{i,t} l_{i,t} - \pi_t \mathbf{1}\{l_{i,t} = 0\}.$$

All other equations remain unchanged and the benefit from question 4 is also still included. The extra tax does not affect the benefit π_t , as only labor income is taxed. The model does not need a new state variable, as the new tax only depends on the year $y_t = t + b$, which is pinned

down by t and b . The tax is also anticipated automatically as we solve by backward induction, which means that workers in years before year 8 already factor the higher future tax into their decisions today, through $\mathbb{E}_t[V_{t+1}]$. Setting $\tau_1 = 0$ results in the model from question 4, and setting both $\tau_1 = 0$ and $\pi_1 = 0$ recovers the baseline model from question 3.

5.b)

To find the τ_1^* that minimizes the effect of the reform on the government budget balance, we minimize the squared gap $(G_3 - G(\theta))^2$ over τ_1 . For each value for τ_1 in the given interval $[0.015, 0.025]$ we solve and simulate the model, computing the total budget $G(\theta) = \sum_i \sum_t G_{i,t}$, and compare it to the baseline G_3 . We use `minimize_scalar` with the bounded method, searching the given interval $[0.015, 0.025]$. The result is:

$$\tau_1^* \approx 0.0236.$$

The extra tax or surtax that pays for the reform is small, being 2.4%. The reason is that the subsidy is targeted, as it is only received by older unemployed/retirees in later years, whereas the surtax is falls on every worker from year 8 and onward, so a small extra tax rate on a wide base raises enough to cover the benefit received by a smaller group. The tax base also holds up well, because $\rho > 1$ the income effect dominates (as we saw in 2.a), why the surtax makes workers work slightly more rather than less.

5.c)

A worker's welfare is measured by the expected discounted lifetime utility at the start of life,

$$W_i = V_0(b_i, k_{i,0}), \quad V_0(b, k_0) = \max \mathbb{E}_0 \left[\sum_{t=0}^{T-1} \beta^t U(c_t, l_t) \right],$$

Which is the model's own value function in the first period, evaluated at worker i 's birth cohort and initial human capital. Higher W_i means that a person is better off, welfare values are negative because of the CRRA form with $\rho > 1$, so a higher (less negative) value means better off. This is exactly what each worker maximizes, so it already contains consumption, leisure and every future period, discounted and with expectations taken over future shocks. It comes from the backward-induction solution, and because it is evaluated at the same birth state $(b_i, k_{i,0})$ under both policies, the comparison is clean as each person is compared with the version of themselves born into the other policy.

In Python we take the solved value function V_0 on the human capital grid, `sol.V[0,b]`, and interpolate it at each worker's initial human capital `k_init`, giving one welfare number per worker.

5.d)

Moving to θ_5^* lowers average welfare in every cohort, and no one is better off (0% for all cohorts). The reform is budget-neutral by construction, yet it makes everyone worse off. The reason comes

Cohort	Avg W (baseline)	Avg W (θ_5^*)	Change	Share better off
0	-30.66	-30.72	-0.062	0%
1	-29.14	-29.39	-0.245	0%
2	-27.70	-27.96	-0.259	0%
3	-26.34	-26.61	-0.269	0%
4	-25.03	-25.31	-0.278	0%

Table 1: Welfare effects of the partly tax-financed reform θ_5^* by birth cohort.

down to who pays and who benefits. We measure welfare at the beginning of life, before anyone knows how their career will turn out. From there, the extra tax is a cost that almost everyone will pay over many years of work. The subsidy is the opposite: only a few people ever collect it. This is because workers build up human capital as they work, so their wages keep rising, and most end up earning far more than the benefit and never choose to retire because of it. So the benefit only helps a small minority. And even that minority gains little, as they retire only when the benefit is barely better than working. So the cost of the tax therefore outweighs the small and uncertain gain from the subsidy for everyone, why no one ends up better off.

Q6

6.a)

What changes between an anticipated and unanticipated reform is only when people find out about it. That timing matters as people are forward-looking, meaning how hard you work today depends on what you believe your future looks like. In the baseline model from question 3, the workers make their decision expecting no reform in the future. In the reform model from question 4 however, the workers make their decisions knowing the reform is coming, so they adjust to it from the beginning of life, even in the years before it actually takes effect. So we already have both sets of beliefs in the two solved models, and we do not need a third one. The anticipated case is simple.

If the reform is anticipated, workers know the reform is coming later on, so they behave as in the reform model from the very first period. That is simply the simulation from question 4.

If the reform is unanticipated, workers expect no reform in the future and behave as in the baseline model until the reform enters as a surprise in year 8. After that they change their beliefs to those in the reform model. So in the simulation, we let people choose hours using the baseline decision rules in every year before year 8. Because the switch is triggered by year and $y = t + b$, different cohorts cross into the reform at different ages. This means that they think they are in the baseline model until year 8. Then at year 8, the reform enters, so we switch them to choose hours based on the reform decision rules. So, we let human capital keep building until the reform enters, so the human capital they have built up by year 8 becomes the state they bring into the reform decision rules.

Switching between two decision models is valid for two reasons. First, both models are solved over the same grid of states (t, b, k) , so the human capital a worker holds at year 8 is a valid input to either model's rules. Second, a worker's optimal choice from year 8 and onward only depends on where they are right now and what the future looks like. It does not depend on what they believed yesterday. This is the recursive (Markov) property of the model: the optimal policy is a function of the current state (t, b, k) alone, with no dependence on history or past beliefs. Once the reform has entered, they expect the reform to be permanent.

So unlike the anticipated reform in question 4, nothing in behavior changes before year 8: the anticipation effect is switched off, and workers only respond once the reform actually arrives.

6.b)

Figure 6b illustrates how the anticipated and unanticipated models differ from the baseline. The labor supply figure shows the gap in average hours between each reform and the baseline. For the first seven years the unanticipated model is zero, which means that the workers are behaving exactly as they would have under no reform. The anticipated line is already below baseline before year 8, since those workers see the subsidy coming and start adjusting in advance. At year 8 both models take a sharp decline as the subsidy arrives, and from there both graphs decline till around year 19. The graphs remain close with the anticipated model sitting slightly below the unanticipated one. After year 19 they increase to around zero and remain flat for the rest of the period.

In the human capital figure, the unanticipated graph stays on baseline until year 7 and then starts to decline around year 8-9, falling in a long, straight line till around year 19, before recovering instantly back to zero. Here too the anticipated line sits a little below the unanticipated one through all years. The contrast with labor supply is, that hours react immediately once the reform has entered, as they are chosen each period. Human capital is accumulated from past work, so it only falls once hours have already been lower for a while.

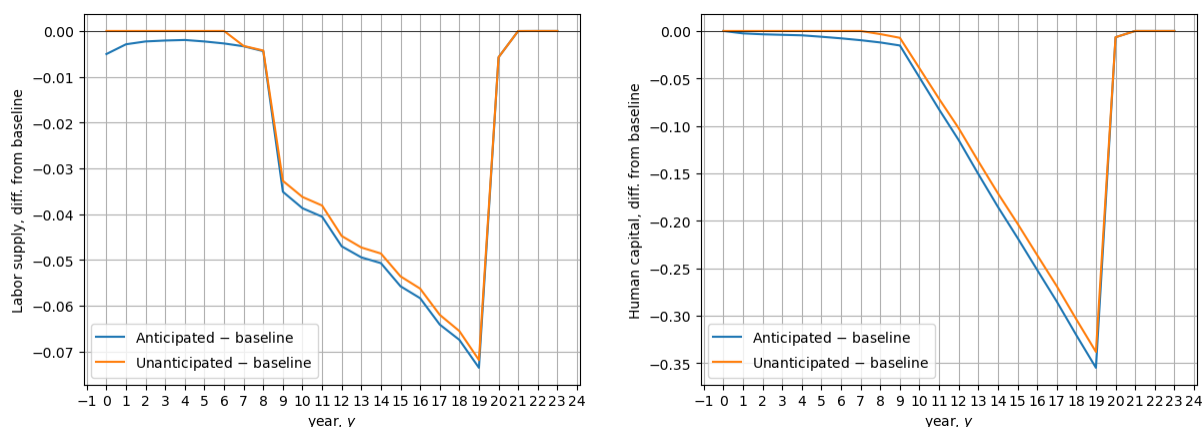


Figure 6b: Difference from baseline: Anticipated vs. unanticipated

6.c)

The behavior in both models is driven by the fact, that the subsidy makes not working attractive for older, low-wage workers, which results in the labor supply falling. As human capital is accumulated by working, the "lost" hours result in a decline in human capital, which lowers the wage, which pulls the hours down even further. This effect is what makes the gap from baseline increase with time (till year 19).

Before year 8 the anticipated workers have already adjusted hours worked, while the unanticipated workers behave exactly as in the baseline. From year 8 and onward both groups face the same rules, why they respond the same way. The only difference is the human capital each group carries into the reform. The unanticipated workers worked as in the baseline right up until the shock happened, so they enter with as much human capital as they would've done in the baseline. This also means that they have slightly higher wages, which keeps them working more hours compared to the anticipated workers. As the anticipated workers adjusted their hours in advance, they enter with less human capital and stay further below the baseline in all periods.

However, the timing of information matters surprisingly little. Most of the labor market response happens after year 8, when both models face the same incentives. Foresight only buys a small head start in lost human capital. Once the low-wage cohorts who actually receive the subsidy are not alive anymore, the only ones left are the high-wage cohorts, who do not receive the subsidy. With no one left who responds to the reform, both lines return to the baseline.

Q7**7.a)**

To capture recessions we now let the aggregate shock include a recession term. Following the corrected exam formulation, we get:

$$\lambda(y_t, r_t) = 1 + \lambda_0 y_t + \lambda_1 r_t,$$

where a recession is denoted by $(r_t = 1)$ and $(r_t = 0)$ denotes a non-recession period. This lowers the wage multiplier by $|\lambda_1|$, and $r_t = 0$ recovers the previous aggregate shock $1 + \lambda_0 y_t$.

The recession indicator r_t becomes a new state variable, so the value function is now $V_t(b, k_t, r_t)$. There are two reasons why r_t becomes a state variable. First, it affects the current payoff as $\lambda(y_t, r_t)$ scales the current wage, such that the optimal choice today depends on whether we are in a recession. Second, it is persistent, as the transition $\Pr(r_{t+1} | r_t)$ depends on r_t , so the worker needs to know today's value of r_t to form the expected continuation value.

The new recursive formulation is, for $1 \leq t < T$,

$$V_t(b, k_t, r_t) = \max_{l_t \in [0,1]} U(c_t, l_t) + \beta \mathbb{E}_t[V_{t+1}(b, k_{t+1}, r_{t+1})]$$

s.t.

$$\begin{aligned} k_{t+1} &= [(1 - \delta)k_t + l_t] \psi_{t+1}, \\ c_t &= (1 - \tau_t) w_t l_t + \pi_t \mathbf{1}\{l_t = 0\}, \\ w_t &= \exp(\alpha_b + \gamma k_t) \lambda(y_t, r_t), \quad \lambda(y_t, r_t) = 1 + \lambda_0 y_t + \lambda_1 r_t, \\ y_t &= t + b. \end{aligned}$$

The expectation now runs over two independent shocks, the individual-specific human-capital shock $\varepsilon_{i,t}$ and the aggregate recession transition r_{t+1} :

$$\mathbb{E}_t[V_{t+1}] = \sum_{r' \in \{0,1\}} \Pr(r_{t+1} = r' \mid r_t) \mathbb{E}_\varepsilon[V_{t+1}(b, k_{t+1}, r')],$$

where $\mathbb{E}_\varepsilon$ is the usual Gauss-Hermite expectation over the human-capital shock. In the terminal period $t = T$ there is no continuation value, so:

$$V_T(b, k_T, r_T) = \max_{l_T \in [0,1]} U(c_T, l_T) \quad \text{s.t.} \quad c_T = (1 - \tau_T) w_T l_T + \pi_T \mathbf{1}\{l_T = 0\}.$$

Setting $\lambda_1 = 0$ (or fixing $r_t \equiv 0$) recovers the baseline model from question 3, so the extension is nested.

7.b)

In Python we added the recession as a new state. We introduced λ_1 as well as the transition matrix given in the exam set. We added the recession indicator r to the aggregate shock, $1 + \lambda_0 y_t + \lambda_1 r_t$. The solution arrays gained an r -dimension, such that the continuation value is now an expectation over both the human-capital shock and next period's recession state, so workers treat recessions as a risk they cannot foresee.

The figure reproduces the recession path used in the simulation. It matches Figure 1 in the exam, recalling that indices start at 0, so a year labelled y here is the exam's year $y + 1$. A recession ($r_t = 1$) occurs seven times over the period, with the rest being non-recession years ($r_t = 0$). The recessions come in three short clusters of two consecutive years plus one isolated year, while longer periods in between are recession-free. In a recession year the aggregate wage multiplier drops by $|\lambda_1|$, which scales every worker's wage down at once, why the marked years here are where we expect to see declines in the year profiles of labor supply and consumption in 7(d).

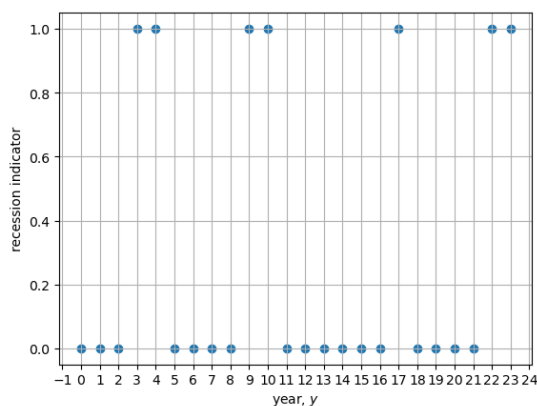


Figure 7a: Recession indicator over time/years

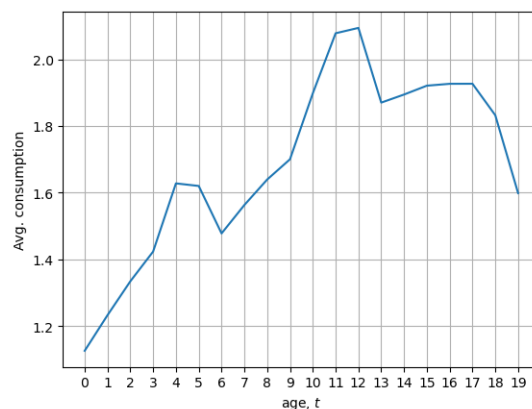
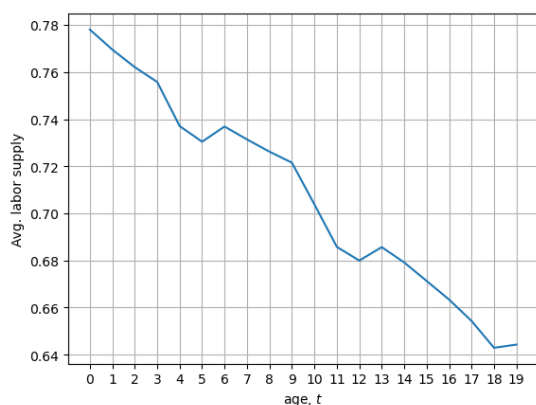
7.c)

The labor-supply and consumption profiles are shaped by human capital, which is accumulated by working. Working hours decline over the life cycle. When young the horizon over which human capital pays off is long, so the return of working is high. As the worker gets older this horizon shrinks and hours fall. Under recessions the labor supply declines a little bit but bounces back under non-recessions.

Consumption is hump-shaped for the same reason. Accumulated human capital raises wages and income early. With age, the income declines as working hours fall near the end of life, and with no savings, consumption simply tracks current income.

The recession-status profile is the telling one. Over age it is not either 0 or 1, but lies between 0 and 0.4. This is because a recession is tied to a calendar year, not an age, so at a given age the five cohorts sit in five different years ($y = t + b$), so the average only measures the share of them in a recession.

We notice slight dips in labor supply and slight increases in consumption at ages where the recession status is at a lower point.



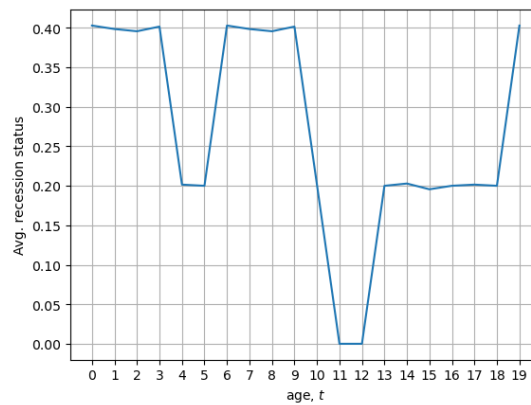
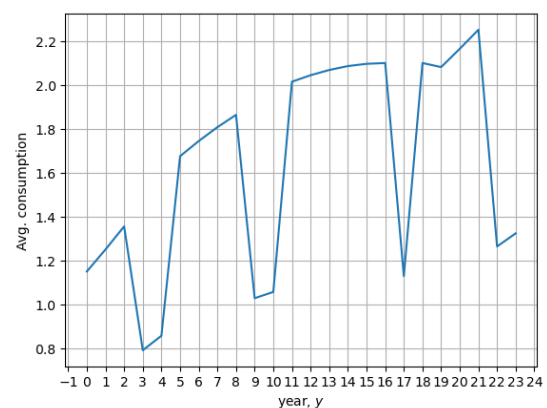
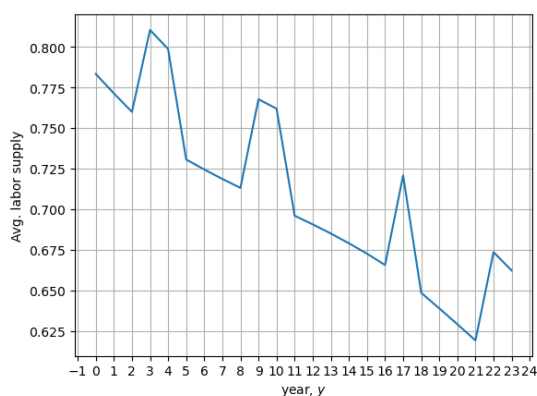


Figure 7c: Labor supply, consumption, recession status as function of age

7.d)

Over time labor supply declines and consumption rises. This is a composition effect from $y = t + b$, since the early years are populated by young workers and the late years by old workers. As older workers have more human capital, they consume more and work less.

However, there are large spikes in the labor supply and large declines in the consumption around the recession years (3+4, 9+10, 17 and 22+23) (remember to plus 1, then it fits with the exam). The two move in opposite directions at exactly the same time, which is because of the wage. Recession scales the wage down, so income declines and with no saving to smooth it, the consumption ends up declining as well. Labor supply moves the other way, as $\rho > 1$ we know the income effect dominates the substitution effect. So when the wage drops, workers put in more hours to partly offset the lost income. So a recession year in this model is characterized by a decline in consumption together with working hours increasing. Both snap back as soon as the recession ends.



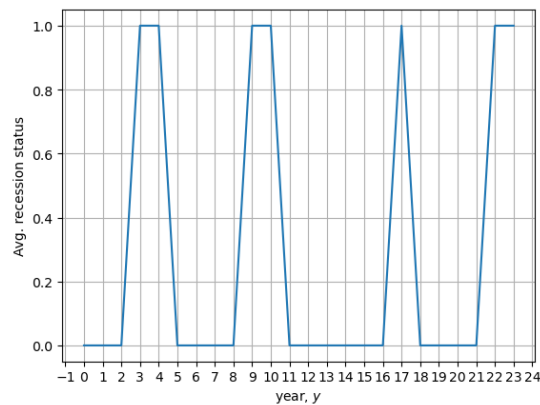


Figure 7d: Labor supply, consumption, recession status as function of year

7.e)

Figure 7c and 7d differ because a recession is an event tied to a calendar year, not to an age. The recession status as a function of year is either 0 or 1, indicating whether a recession is present in the current year or not. When the recession status is a function of age, the average recession status lies between 0 and 0.4 rather than being 0 or 1. This is because at a given age the five cohorts are in five different years ($y = t + b$), and only some of these are recession years, so the average only measures the share of cohorts in a recession at that age.

The same difference shows up in labor supply and consumption. As a function of year, the recessions appear as sharp increases in consumption and spikes in labor supply that line up exactly with the true recession years. As a function of age, these movements are averaged across cohorts that sit in different years, so they nearly disappear and the profiles mostly show the smooth life cycle from human capital. The behavior behind the two figures is the same, the only difference is that organizing the data by year lines up the workers who share the same recession, while organizing by age mixes cohorts living through different years and averages the recession away.